# Optimization Theory

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# Course Info

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# Optimization

- General optimization/programming problem in standard form

$$
\begin{aligned}
&\min_x f_0(x) \\
&s.t.\ f_i(x) \leq 0 \qquad i = 1, \cdots, m \\
&\qquad h_i(x) = 0 \qquad i = 1, \cdots, p
\end{aligned}
$$

  - $x$ is the optimization variable
  - $f_0$ is the objective function
  - $f_i, i = 1, \cdots, m$ are inequality constraint functions
  - $h_i, i = 1, \cdots, p$ are equality constraint functions
- Goal: Find an optimal solution $x^*$ that minimizes $f_0$ while satisfying all the constraints.

# Applications

- Communications technology
  - ML detection
  - Resource allocation
  - Network flow
  - TCP protocol
- Information processing
  - Filter design
  - Image compression
  - Data classifier
- More applications in our life
  - Delivery service
  - Taxi scheduling
  - Portfolio optimization

# Power Allocation Example

- Goal: Design the power allocation problem to maximize the total throughput over $N$ parallel channels:

$$\max_{\mathbf{P}} \quad \mathbb{E}\left[\sum_{i=1}^{N} \log\left(1+\frac{p_i g_i}{n_0}\right)\right]$$

$$s.t. \quad \mathbb{E}\left[\sum_{i=1}^{N} p_i\right] \leq \overline{P}$$

$$\sum_{i=1}^{N} p_i h_i \leq \overline{Q}$$

$$p_i \geq 0 \qquad \forall i = 1, \cdots, N$$

  - where $p_i$ is the transmit power, $g_i$ and $h_i$ are the channel gains of the transmission and interferring links, $n_0$ is the thermal noise, $\overline{P}$ is the time-averaging total power constraint and $\overline{Q}$ is the instantenous interference limitation.

# Solving Optimization Problem

- General optimization problem
  - Very difficult to solve
  - Local optimization: fast but provide no information about distance to global optimum
  - Global optimization: worst-case complexity grows exponentially with problem size
- We mainly focus on tractable optimization problems
  - Linear optimization
  - Convex optimization
  - Stochastic optimization

# Solving Optimization Problem

- Linear Optimization

$$\begin{aligned} \min_{x} \quad & c^T x \\ s.t. \quad & a_i^T x \leq b_i \qquad \forall i \end{aligned}$$

- Convex Optimization

$$\begin{aligned} \min_{x} \quad & f_0(x) \\ s.t. \quad & f_i(x) \leq b_i \qquad \forall i \end{aligned}$$

- Stochastic Optimization

$$\begin{aligned} \min_{x} \quad & \mathbb{E}[f_0(x, \omega)] \\ s.t. \quad & f_i(x, \omega) \leq b_i \qquad \forall i \end{aligned}$$

# Solving Optimization Problem

- How to solve the power allocation problem?

$$
\begin{aligned}
\max_{\mathbf{P}} \quad & \mathbb{E}\left[\sum_{i=1}^{N} \log\left(1+\frac{p_i g_i}{n_0}\right)\right] \\
s.t. \quad & \mathbb{E}\left[\sum_{i=1}^{N} p_i\right] \leq \overline{P} \\
& \sum_{i=1}^{N} p_i h_i \leq \overline{Q} \\
& p_i \geq 0 \qquad \forall i = 1, \cdots, N
\end{aligned}
$$

# Solving Optimization Problem

- The answer is: It depends on how much you know about optimization.
- If you don't know anything, then you just take a heuristic guess like using a uniform power $p_i = \overline{P}/N$ or a proportional power $p_i = \frac{g_i}{\sum_{i=1}^{N} g_i}\overline{P}$.
- If you know about linear optimization, then approximate the problem as

$$
\begin{aligned}
\max_{\mathbf{P}} \quad & \sum_{i=1}^{N} p_i g_i \\
s.t. \quad & \sum_{i=1}^{N} p_i \leq \overline{P} \\
& \sum_{i=1}^{N} p_i h_i \leq \overline{Q} \\
& p_i \geq 0 \qquad \forall i = 1, \cdots, N
\end{aligned}
$$

# Solving Optimization Problem

- If you know about convex optimization, then approximate the problem as

$$
\begin{aligned}
\max_{\mathbf{P}} \quad & \sum_{i=1}^{N} \log\left(1+\frac{p_i g_i}{n_0}\right) \\
s.t. \quad & \sum_{i=1}^{N} p_i \leq \overline{P} \\
& \sum_{i=1}^{N} p_i h_i \leq \overline{Q} \\
& p_i \geq 0 \qquad \forall i = 1, \cdots, N
\end{aligned}
$$

- If you know about stochastic optimization, you can solve the original problem.

# Course Goal

- Introduce basic concepts/principles/approaches in optimization theory
- Recognize/formulate optimization problems as solvable styles
- Learn to solve optimization problems using software
- Illustrate its applications in communications and signal processing
- Provide useful tools for your own research

# Syllabus

1. Linear Optimization
2. Convex Optimization - Basic Concepts
3. Duality Theory
4. Solving Convex Optimization Problems
5. Decomposition
6. Combinatorial Optimization and Relaxation
7. Stochastic Optimization

# Reference Books

- S. Boyd, L. Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004
  http://stanford.edu/~boyd/cvxbook/
- Further Reading
  - D. P. Palomar and Y. C. Eldar, Eds., *Convex Optimization in Signal Processing and Communications*, Cambridge University Press, 2009
  - M. Chiang, S. H. Low, Z. Luo, N. B. Shroff, W. Yu, eds., special issue on Nonlinear Optimization of Communication Systems, *IEEE Journal on Selected Areas of Communication*, vol. 24, no. 8, 2006

# Reference Courses

Acknowledgement to the follows professors for material for our course.

- Prof. Stephen Boyd, *Convex Optimization*, Standford University
- Prof. Daniel P. Palomar, *Convex Optimization*, Hong Kong University of Science and Technology
- Prof. Vincent K. N. Lau, *Advanced Stochastic Optimizations for Wireless Systems*, Hong Kong University of Science and Technology
- Prof. Mung Chiang, *Optimization of Communication Systems*, Princeton University

# Grading

- Class paticipation: 10% (Let TA know if you have any problem)
- Homework: 30% (5 assignments)
- Final Project: 60%